

2016.0 RANGE ROVER (LG), 501-11

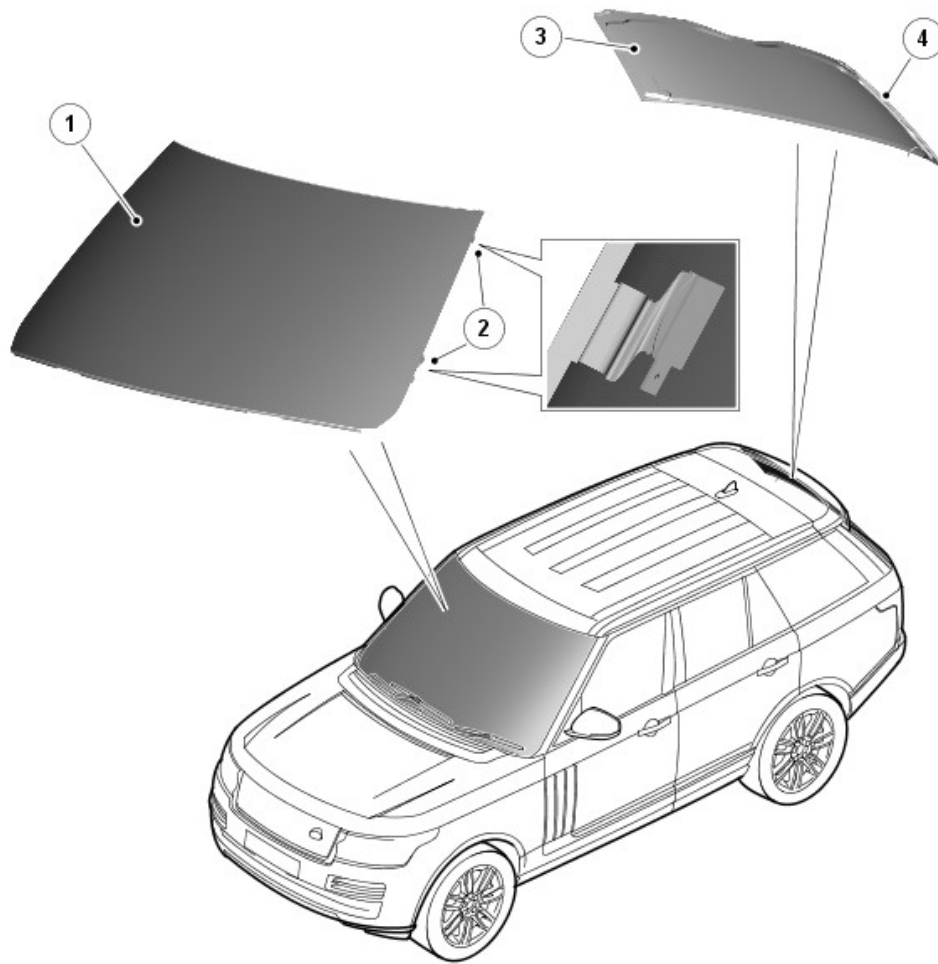
GLASS, FRAMES AND MECHANISMS

DESCRIPTION AND OPERATION

COMPONENT LOCATION - FIXED GLASS

NOTE:

Standard wheelbase vehicle shown; long wheelbase vehicles similar.



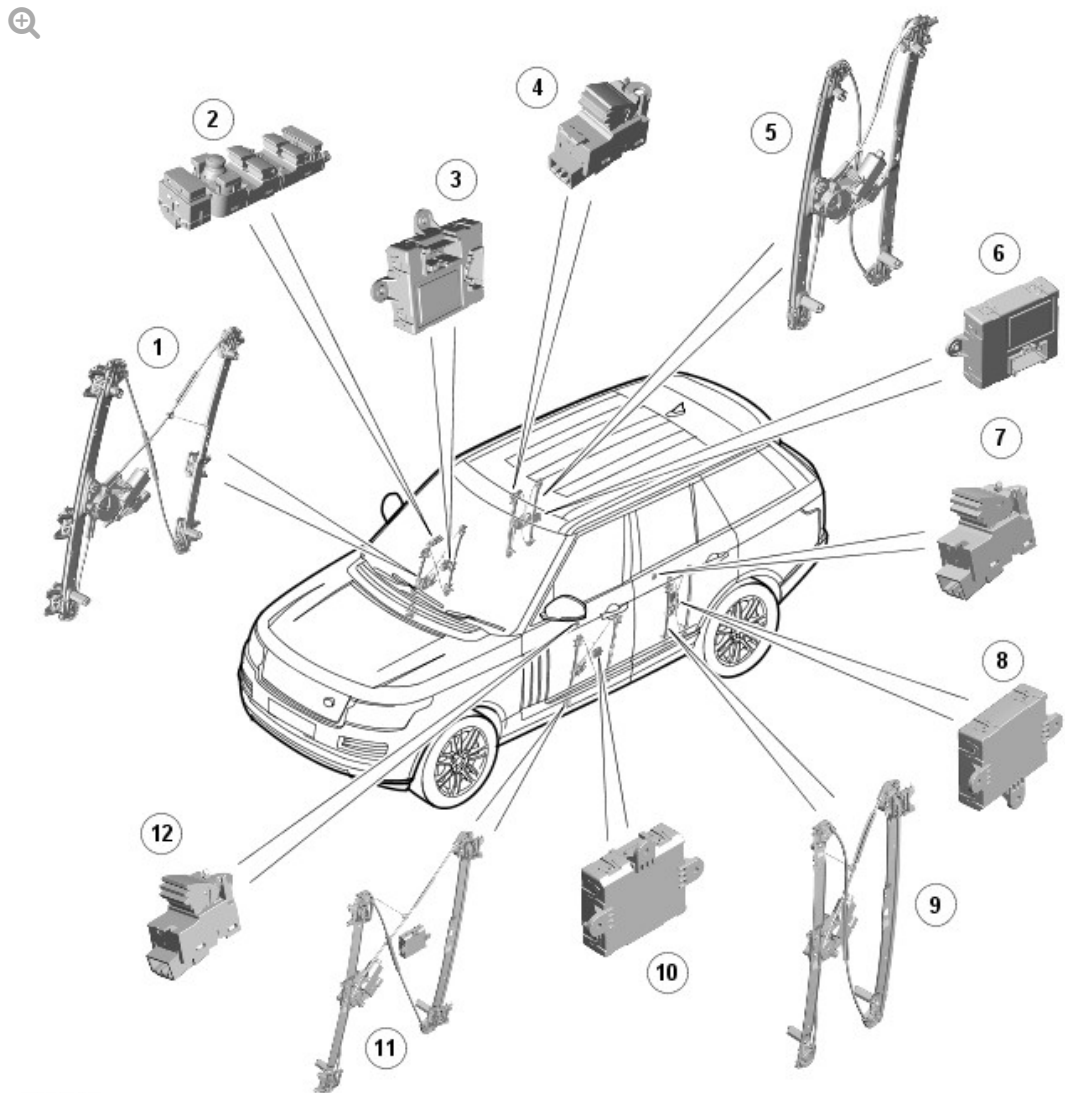
E149173

ITEM	DESCRIPTION
1	Windshield
2	Heated windshield connectors
3	Rear window
4	Finisher

COMPONENT LOCATION - DOOR GLASS AND MECHANISMS

NOTE:

Standard wheelbase vehicle shown; long wheelbase vehicles similar.

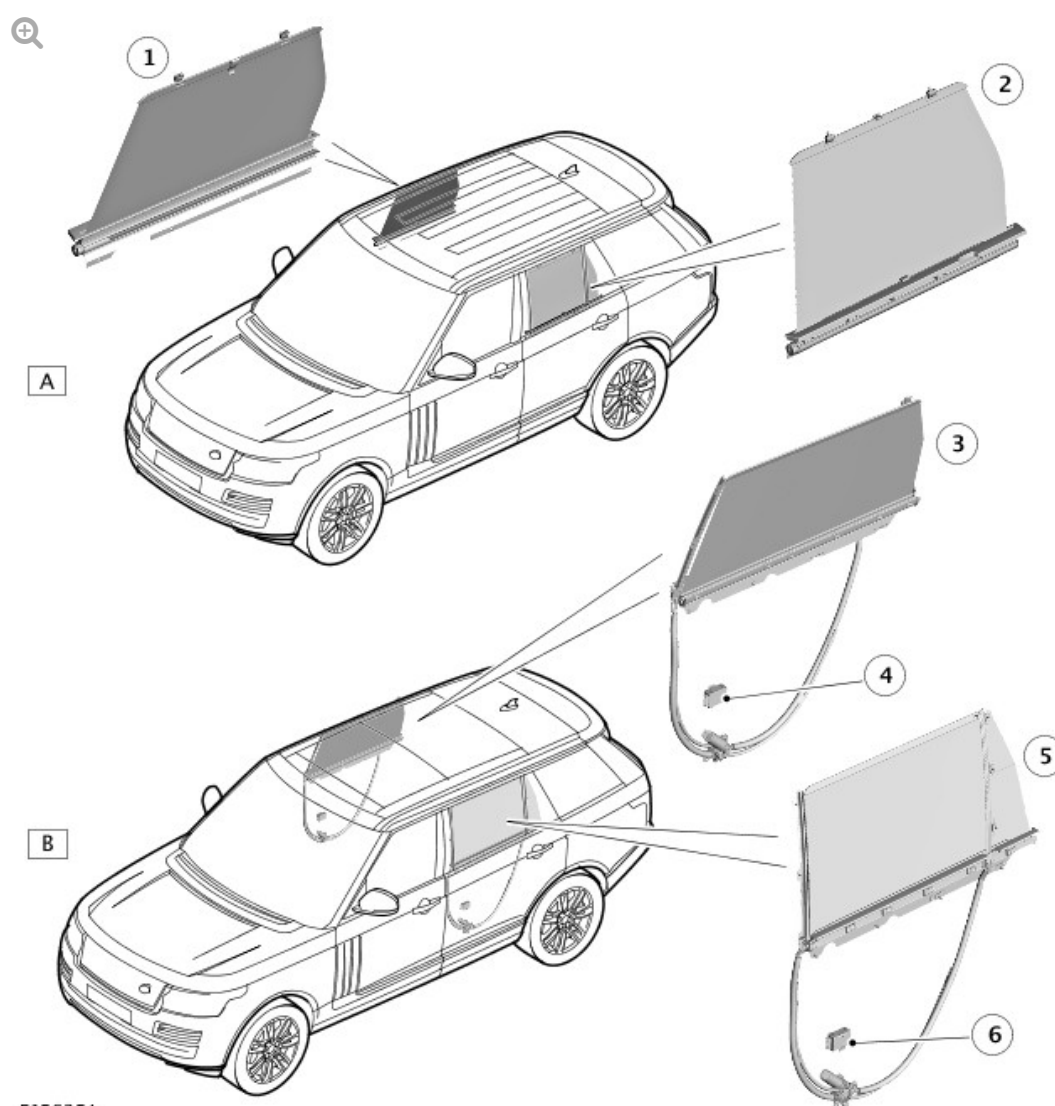


E149174

ITEM	DESCRIPTION
1	Driver door regulator and motor
2	Driver door switchpack
3	Driver door module
4	Rear right door switchpack
5	Rear right door regulator and motor
6	Rear door module

7	Rear left door switchpack
8	Rear door module
9	Rear left door regulator and motor
10	Passenger door module
11	Front left door regulator and motor
12	Front left door switchpack

COMPONENT LOCATION - DOOR BLINDS



E156354

ITEM	DESCRIPTION
1	Right door blind

2	Door blind module
3	Left door blind
4	Door blind module

OVERVIEW

DOOR WINDOW MECHANISMS

Operation of the window lift system is controlled by the CJB (central junction box) in conjunction with the driver door module and the front passenger door module. The door modules are mounted in the door of their respective windows. Door modules are also mounted in both rear doors, and communicate with the front door modules over the LIN (local interconnect network) bus.

The window lift system features one shot up and down operation, plus an anti-trap feature for all door windows.

If the battery is disconnected, the window lift system will require calibrating on battery re-connection. To calibrate the system the following routine must be carried out.

With the ignition in power mode 6:

- Fully close the door glass and release the switch.
- Re-apply the switch to close the door glass and hold for approximately 2 seconds.
- Fully open the door glass and release the switch.
- Re-apply the switch to open the door glass and hold for approximately 2 seconds.
- Repeat for all remaining windows.

FIXED GLASS

The laminated windshield is bonded and sealed to the body aperture using

PU (polyurethane) adhesive.

The toughened rear window is also bonded and sealed to the tailgate aperture using PU adhesive. A single heating element is printed on the inner surface of the rear window.

DOOR BLINDS

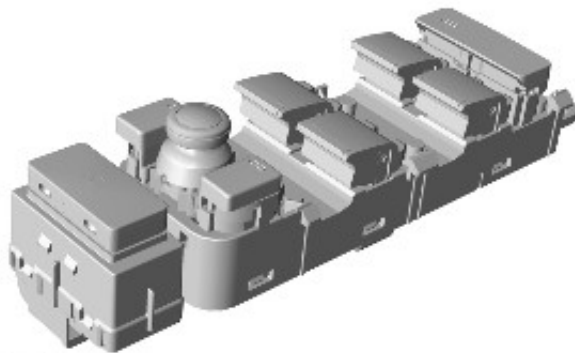
Depending on model and market specification, vehicles may incorporate electric blinds on the rear doors.

Each door blind has an electric motor that drives Bowden cables attached to a roller blind. A door blind module and the door switchpacks control the motor to raise and lower the blind.

DESCRIPTION

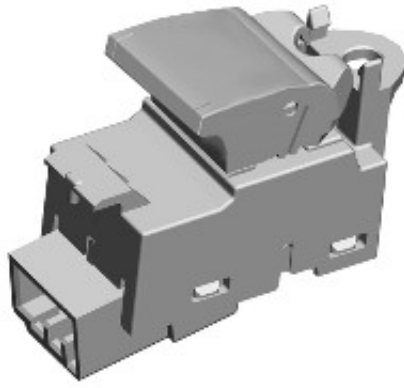
WINDOW SWITCHES

Driver Door Switchpack



E149175

Passenger Door Switchpack



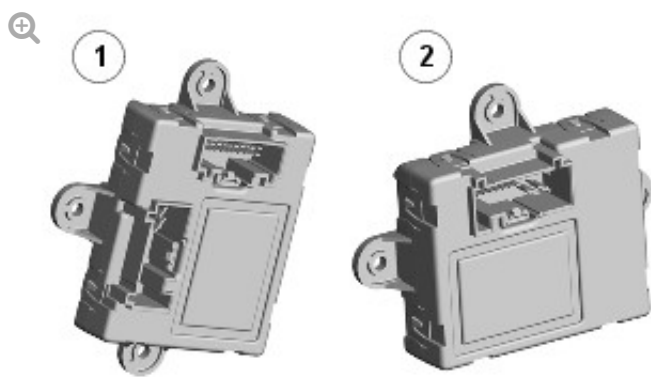
E156355

Individual window switches are installed in each of the three passenger doors. Window switches for all of the windows and a rear window isolation switch are installed in the driver door switchpack in the top surface of the door trim.

All window switches are of the non-latching rocker type. The switches have two switching positions in each direction, inch up/down and one-shot up/down. Operating the switch to the second detent position will activate the one-shot feature.

The driver door switchpack is powered by a permanent battery feed from the CJB. When the switches in the driver door switchpack are used, the switchpack translates the switch movement into a LIN bus message. A LIN bus connects the driver door switchpack to the driver door module and the driver side RDM (rear door module). Each passenger window switch is hardwired into circuits with the related door module.

DOOR MODULES



E144993

ITEM	DESCRIPTION
1	Front door module
2	Rear door module

In addition to controlling the window lift system, the front door modules also control:

- Door locking.
For additional information, refer to: [Handles, Locks, Latches and Entry Systems](#) (501-14 Handles, Locks, Latches and Entry Systems, Description and Operation).
- Door mirror adjustment and heating.
For additional information, refer to: [Rear View Mirrors](#) (501-09 Rear View Mirrors, Description and Operation).
- Door mirror lamps.
For additional information, refer to: [Interior Lighting](#) (417-02 Interior Lighting, Description and Operation).
- Mirror proximity cameras.
For additional information, refer to: [Parking Aid](#) (413-13 Parking Aid, Description and Operation).

The rear door modules control window lift, door locking and (where fitted) the door blind via the door blind module.

The door modules are color coded for position on the vehicle. The driver door module and both rear door modules are black. The passenger door module is gray. The electrical wiring harness connectors are also color coded to match the door modules.

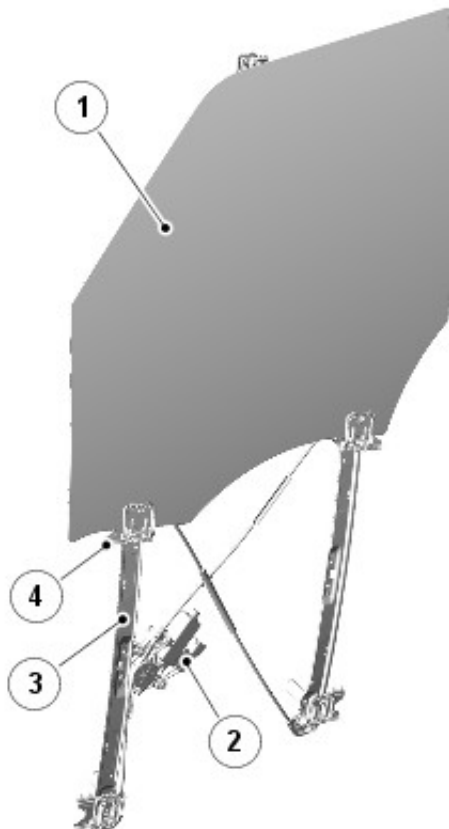
NOTE:

Both rear door modules are identical.

If a front door module develops a fault, a DTC (diagnostic trouble code) is stored in its memory, DTC's can be read using the Land Rover approved diagnostic equipment. If either of the rear door modules develops a fault, the DTC is stored in the respective front door module.

WINDOW REGULATORS

Front Window Regulators



E149177

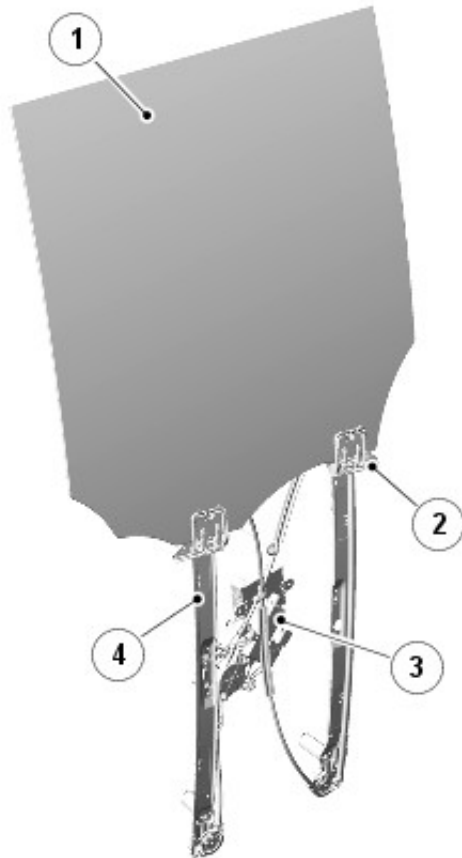
ITEM	DESCRIPTION
1	Window Glass
2	Window motor
3	Mounting frame
4	Glass carrier

The front window regulator and motor is supplied as an assembly and is handed. Each assembly comprises a front and rear runner, a continuous cable and a motor.

The runners are secured in the door frame with four bolts. The door glass is located in two carriers, which are located in tracks in the runners. The glass is retained in friction pads in each carrier and secured with clamps.

Each carrier is attached to the cable which, in turn, is attached to a drum driven by the motor. When the motor is operated the drum pulls the cable in the required direction to raise or lower the glass.

Rear Window Regulators



E149178

ITEM		DESCRIPTION
1		Window Glass
2		Glass carrier
3		Window motor
4		Mounting frame

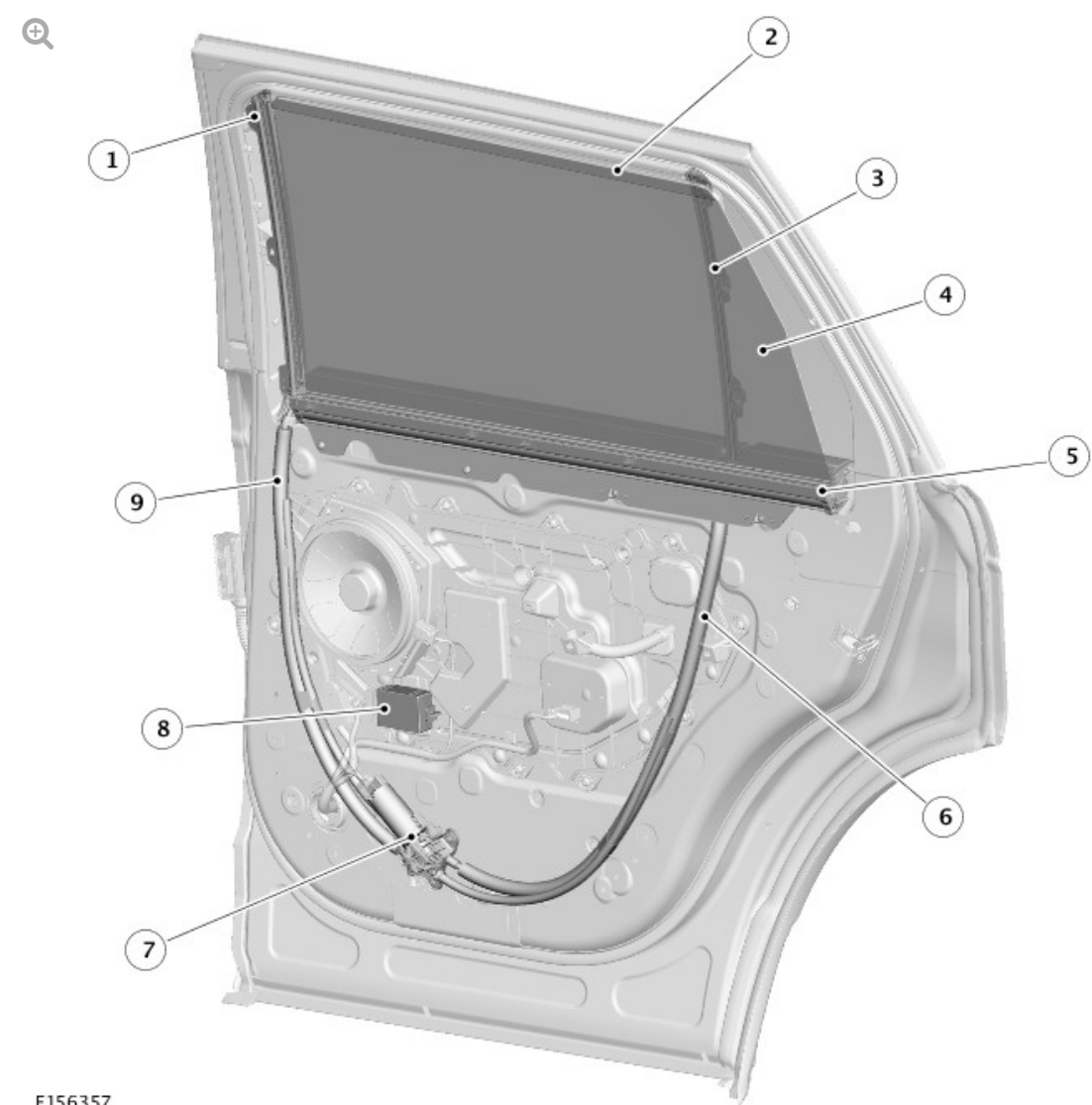
The rear window regulator and motor is supplied as an assembly and is handed. Each assembly comprises a front and rear runner, a continuous cable and a motor.

The runners are secured in the door frame with four bolts. The door glass is located in two carriers, which are located in tracks in the runners. The glass is retained in friction pads in each carrier and secured with clamps.

Each carrier is attached to the cable which, in turn, is attached to a drum driven by the motor. When the motor is operated the drum pulls the cable

in the required direction to raise or lower the glass.

DOOR BLINDS



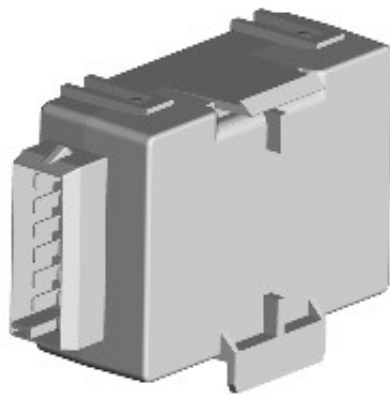
E156357

ITEM	DESCRIPTION
1	Front guide rail
2	Top bar
3	Rear guide rail
4	Roller blind
5	Roller blind housing
6	Rear Bowden cable
7	Door blind motor
8	Door blind module

Each door blind has a motor and a roller blind housing, attached to the inner door panel, and guide rails attached to the B pillar finisher and the glass divider bar of the door. The ends of a top bar, attached to the roller blind, run in slots in the guide rails. Bowden cables connect the ends of the top bar to the motor.

The motor consists of a reversible DC (direct current) motor attached to a worm drive gearbox. The two Bowden cables locate in the gearbox and are driven by the worm gear. The door harness connects the motor to the door blind module, which provides power and ground connections for the motor and monitors operation of the door blind using inputs from two Hall effect sensors in the motor.

DOOR BLIND MODULES



E156358

Each door blind is operated by a door blind module clipped onto the closing panel of the related door. In addition to connections with the door blind motor, each door blind module has a permanent power feed from the EJB (engine junction box) and a LIN bus connection with the front and rear door modules on the related side of the vehicle.

OPERATION

DOOR WINDOWS

The electric windows will operate in power modes 6 (ignition on) and 7 (engine running), and for five minutes after the ignition is switched off provided none of the doors are opened.

When a window open or close selection is made, the related door module supplies power to the window motor to drive it in the appropriate direction. In the inch mode, the motor stops when the switch is released or the window reaches the end of its travel. In the one-shot mode, the motor stops when the switch is operated again (either up or down) or the window reaches the end of its travel.

When the passenger window switches are used, they produce an open or close request by completing a circuit with the related door module.

When the driver window switches are used, the driver door switchpack outputs a request message for the appropriate door module on the LIN bus. If the message is for a door module on the opposite side of the vehicle to the driver, the DDM (driver door module) relays the message to the PDM (passenger door module) on the medium speed CAN (controller area network) body systems. If necessary, the PDM then sends a LIN bus message to the RDM on its side of the vehicle.

If any of the passenger windows have opposing up and down requests from two separate switches, for example, a passenger window switch and the related window switch on the driver door switchpack, then the operation of that window will cease, until one or both of the switches are released.

When the rear window isolator switch is engaged, the switch tell-tale is illuminated and the rear door modules ignore requests from their related passenger window switches.

Global opening and closing requests are output from the CJB on the medium speed CAN body systems. The DDM and the PDM then relay the request to their respective RDM.

ONE-SHOT WINDOW OPERATION RESET

If the battery is disconnected or discharged, or the power supply to a door module is interrupted, one-shot window operation is disabled until the window position is re-established by the affected door module(s). To reset one-shot window operation:

- 1** Close the window fully.
- 2** Release the switch, then pull up and hold the switch for one second.
- 3** Repeat the procedure for the other windows.

ANTI-TRAP PROTECTION

The anti-trap feature is incorporated for all of the door windows in both the inching and one-shot modes. If the anti-trap function is activated while a window is closing, the window motor reverses for 0.5 second.

Each window motor has a Hall sensor to enable the related door module to monitor the motor speed. If the motor speed decreases below a set threshold, indicating an obstruction, the power feed to the motor is reversed so the window goes back down.

If it is still necessary to raise the window, the anti-trap protection can be overridden by attempting to close the window at intervals of less than 10 seconds. On the third attempt the window will move up with increased force to try and overcome the obstruction. If the obstruction cannot be overcome, one-shot operation is disabled.

DOOR BLINDS

The door blinds only operate when the related rear door window is closed. With the window closed, the door blind will operate while the ignition is on and for 5 minutes after the ignition is switched off provided none of the doors are opened. The door blinds can only be fully raised or fully lowered, there is no intermediate position.

The position of the rear door windows is transmitted to each door blind module in a LIN bus message from the related RDM.

The door blinds are operated by either the rear door switchpacks or the rear window switches in the driver door switchpack. Pulling a switch raises the related door blind. Pressing a switch lowers the related door blind. When a switch selection is made, a corresponding LIN message is transmitted to the related door blind module from the RDM (if a rear door switchpack is used) or the driver door switchpack (if a rear window switch in the driver door switchpack is used) via the corresponding RDM. Provided the door window is closed and the selection is valid, the door blind module then energizes the motor to drive the door blind in the appropriate direction.

When the isolator switch in the driver door switchpack is engaged, operation of the door blinds with the rear door switchpacks is inhibited. Operation remains possible with the rear switches on the driver door switchpack.

ANTI-TRAP PROTECTION

If a door blind is restricted for any reason while being raised, the blind will stop and return to the lowered position.

THERMAL PROTECTION

Deactivation will occur if a door blind is operated more than 20 up/down cycles within a minute. The door blind will not operate again for approximately 60 seconds.

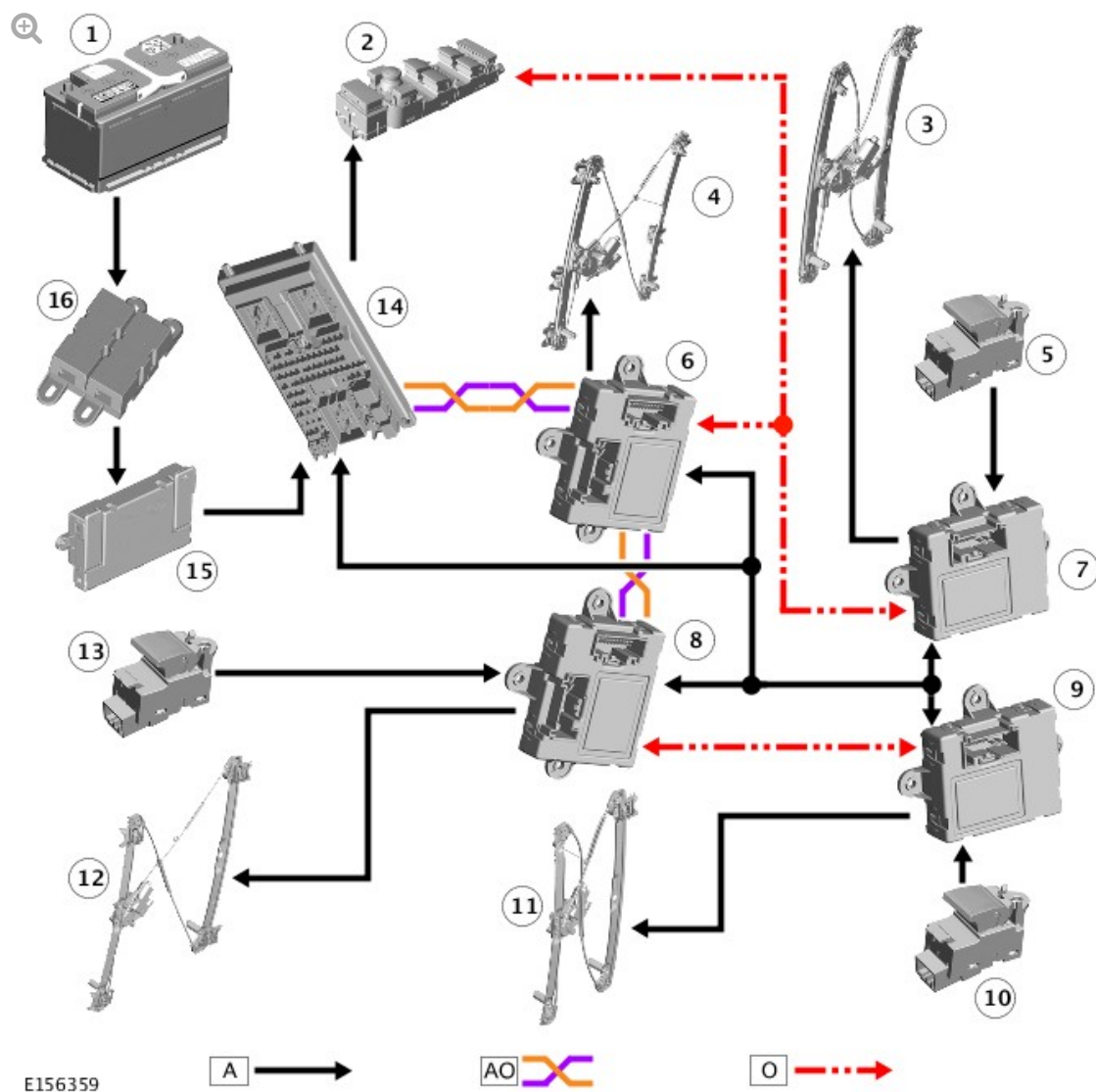
COLD WEATHER PERFORMANCE

Operating performance of the door blinds is reduced if the temperature inside the vehicle is less than -10°C (14°F).

NOTE:

Testing of the door blind system should be conducted in temperatures above 0°C (32°F).

CONTROL DIAGRAM - DOOR WINDOWS

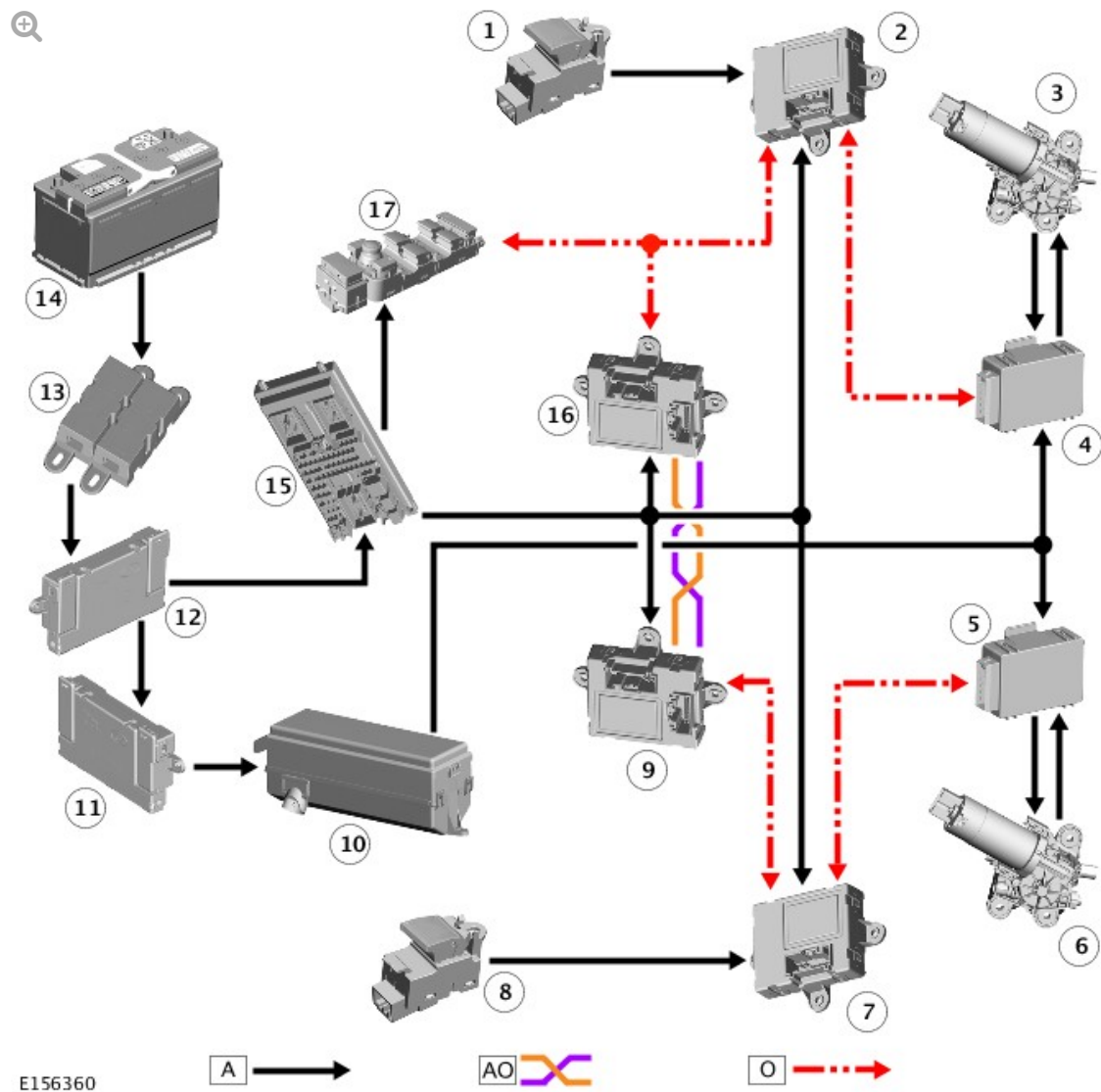


A = HARDWIRED; AO = MEDIUM SPEED CAN BODY SYSTEMS; O = LIN BUS.

ITEM	DESCRIPTION
1	Battery
2	Driver door switchpack
3	Rear door glass motor
4	Driver door glass motor
5	Rear door switchpack
6	Driver door module
7	Rear door module

8	Passenger door module
9	Rear door module
10	Rear door switchpack
11	Rear door glass motor
12	Passenger door glass motor
13	Passenger door switchpack
14	Central junction box
15	Battery junction box
16	Battery junction box 2

CONTROL DIAGRAM - DOOR BLINDS



A = HARDWIRED; AO = MEDIUM SPEED CAN BODY SYSTEMS; O = LIN BUS.

ITEM	DESCRIPTION
1	Rear door switchpack
2	Rear door module
3	Door blind motor
4	Door blind module
5	Door blind module
6	Door blind motor
7	Rear door module
8	Rear door switchpack
9	Passenger door module
10	Engine junction box
11	Auxiliary junction box
12	Battery junction box
13	Battery junction box 2
14	Battery
15	Central junction box
16	Driver door module
17	Driver door switchpack